

Chulwoo Kim

professor

基本信息

姓名: Chulwoo Kim

所属院校: 高丽大学

地区: South Korea

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

低功耗电路与能量管理

模拟与数模混合电路

超大规模集成电路设计

数字集成电路设计

混合信号集成技术

微电子器件与制造

模拟与射频IC设计

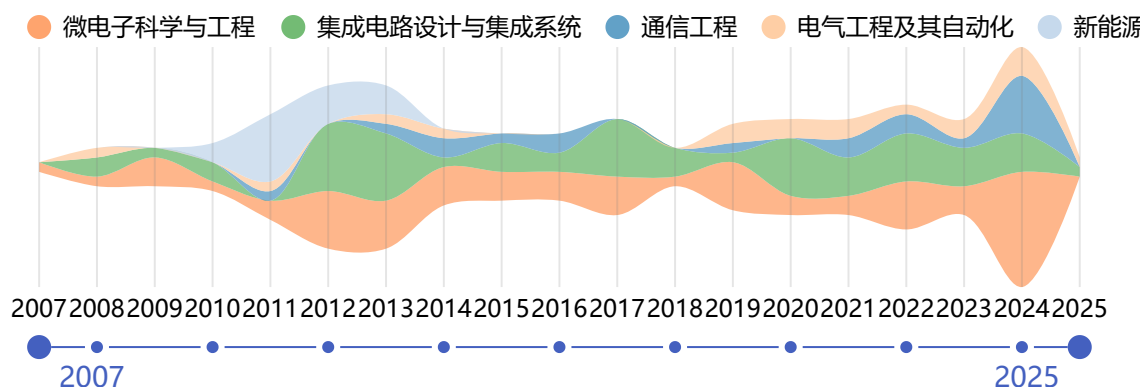
射频集成电路

存储器设计与架构

个人简介

Chulwoo Kim is a professor at the School of Electrical Engineering. His research interests include low power, energy engineering, energy harvesting, CMOS process, etc. He has an h-index of 19 (from 1996 - 2024) and 27 citations. With a total of 259 research outputs, including 155 articles, 81 conference contributions, 13 conference articles, etc. In 2024, he had multiple publications in top-tier IEEE journals, such as IEEE Journal of Solid-State Circuits and IEEE Transactions on Circuits and Systems II: Express Briefs. He also had media coverage in a COVID-19 study in 2020.

近年发表的论文



A 10 Gb/s Single-Ended Receiver Using Time Gap Sense Amplifier for Next-Generation HBM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Taehwan; Sim, Changmin; Park, Seungwoo; Park, Jinwoo; Kim, Seongcheol; Shin, Hwaseok; So, Junseob; Lee, Seon-Been; Kwon, Youngwook; Kim, Chulwoo

被引用次数: 0

2025

An Asynchronous Geometric Sequence Array-Based Binary Search Digital Low-Dropout Regulator for $O(\log n)$ -Complexity Coarse Searching Operations

IEEE TRANSACTIONS ON POWER ELECTRONICS

Kim, Hyunjin; Park, Taehyeong; Kim, Seokjin; Han, Seokhee; Jeong, Mingi; Kim, Jaekyun; Park, Inho; Kim, Chulwoo

被引用次数: 0

2025

A Fully Integrated Nine-Ratio Switched-Capacitor Converter With Overlapped-Conversion-Ratio Modulation for IoT Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hyunjin; Park, Taehyeong; Park, Inho; Park, Changhun; Kim, Seokjin; Han, Seokhee; Jeong, Junwon; Kim, Chulwoo

被引用次数: 0

2024

Single-Ended PAM-4 Transmitters With Data Bus Inversion and ZQ Calibration for High-Speed Memory Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Seongcheol; Sim, Jincheol; Choi, Yoonjae; Choi, Jonghyuck; Kwon, Youngwook; Park, Seungwoo; Sim, Changmin; So, Junseob; Park, Taehyeong; Kim, Chulwoo

被引用次数: 2

2024

A 0.458-pJ/bit 24-Gb/s/pin Capacitively Driven PAM-4 Transceiver With PAM-Based Crosstalk Cancellation for High-Density Die-to-Die Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Seongcheol; Sim, Changmin; Sim, Jincheol; Choi, Jonghyuck; Kwon, Youngwook; Park, Seungwoo; So, Junseob; Shin, Hwaseok; Lee, Seon-Been; Kim, Chulwoo

被引用次数: 2

2024

A Single-Ended NRZ Receiver With Gain-Enhanced Active-Inductive CTLE and Reference-Selection DFE for Memory Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Jonghyuck; Choi, Yoonjae; Sim, Jincheol; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Sim, Changmin; Kim, Chulwoo

被引用次数: 0

2024

A Wireline Transceiver With 3-bit per Symbol Using Common-Mode NRZ and Differential-Mode PAM-4 Signaling Techniques

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sim, Jincheol; Choi, Jonghyuck; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Sim, Changmin; Shin, Hwaseok; So, Junseob; Lee, Seonbeen; Kim, Chulwoo

被引用次数: 1

2024

A 10-Gb/s Wireline Receiver Using Linear Baud-Rate CDR and Analog Equalizer for Free Space Optical Communication Over 10-and 100-m Distances

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sim, Jincheol; Sim, Changmin; Park, Hyunsu; Choi, Yoonjae; Choi, Jonghyuck; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Kim, Jong-Min; Lee, Ju-Hyung; Ko, Young-Chai; Kim, Chulwoo

被引用次数: 0

2024

A Four-Phase Time-Based Switched-Capacitor LDO With 13-ns Settling Time at 0.5-V Input for Energy-Efficient Computing in SoC Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hyunjin; Park, Changhun; Park, Inho; Park, Taehyeong; Park, Seungwoo; Kim, Chulwoo

被引用次数: 3

2024

A Time-Based Direct MPPT Technique for Low-Power Photovoltaic Energy Harvesting

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Maeng, Junyoung; Jeong, Junwon; Park, Inho; Shim, Minseob; Kim, Chulwoo

被引用次数: 5

2024

A 50 Gb/s PAM-4 Transceiver With High-Swing Driver, Dual-Loop Analog Equalizer, and Integrator-Based Baud-Rate Linear CDR for Short-Reach Links

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Sim, Jincheol; Sim, Changmin; Choi, Jonghyuck; Park, Seungwoo; Kim, Seongcheol; Kim, Chulwoo

被引用次数: 0

2024

A 25-Gb/s Single-Ended PAM-4 Transmitter With iPWM-Based FFE and RLM-Matched Voltage-Mode Driver for High-Speed Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Choi, Yoonjae; Sim, Changmin; Choi, Jonghyuck; Sim, Jincheol; Park, Hyunsu; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Kim, Chulwoo

被引用次数: 0

2024

A Fully Integrated Dual-Output Continuously Scalable-Conversion-Ratio SC Converter for Battery-Powered IoT Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Park, Taehyeong; Kim, Hyunjin; Jeong, Mingi; Park, Inho; Kim, Chulwoo

被引用次数: 0

2024

A 0.45 pJ/b 24 Gb/s NRZ Receiver Data-Path Using Half-Baud-Rate Duobinary Sampling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Park, Seungwoo; Choi, Yoonjae; Choi, Jonghyuck; Sim, Jincheol; Kwon, Youngwook; Sim, Changmin; Kim, Seongcheol; Kim, Chulwoo

被引用次数: 0

2024

A 101.6-dB-SNDR Fully Dynamic Zoom ADC Using Miller-Compensated Floating Inverter Amplifiers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Yohan; Lee, Woojin; Park, Sooho; Kim, Changjoo; Jung, Hyundo; Kim, Chulwoo

被引用次数: 0

2024

A 13-Gb/s Single-Ended NRZ Receiver With 1-Sample Per 2-UI Using Data Edge Sampling for Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kang, Hyoshin; Choi, Yoonjae; Sim, Jincheol; Choi, Jonghyuck; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Sim, Changmin; Kim, Chulwoo

被引用次数: 0

2024

A 28-Gb/s Single-Ended PAM-4 Transceiver With Active-Inductor Equalizer and Amplitude-Detection LSB Decoder for Memory Interfaces

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Shin, Hwaseok; Kang, Hyoshin; Choi, Yoonjae; Sim, Jincheol; Choi, Jonghyuck; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Sim, Changmin; So, Junseob; Kim, Taehwan; Kim, Chulwoo

被引用次数: 0

2024

A Tri-Mode Reconfigurable DC-DC Converter With Photovoltaic Energy Harvesting for Miniature IoT Batteries

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Maeng, Junyoung; Park, Inho; Jeon, Jinwoo; Kim, Hyunjin; Lee, Hoi; Kim, Chulwoo

被引用次数: 0

2024

A 52-Gb/s Low-Power PAM-4 Baud-Rate CDR Using Pattern-Based Phase Detector for Short-Reach Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Seungwoo; Choi, Yoonjae; Sim, Jincheol; Choi, Jonghyuck; Park, Hyunsu; Kwon, Youngwook; Sim, Changmin; Kim, Seongcheol; Kim, Chulwoo

被引用次数: 0

2024

A 33-Gb/s/Pin 1.09-pJ/Bit Single-Ended PAM-3 Transceiver With Ground-Referenced Signaling and Time-Domain Decision Technique for Multi-Chip Module Memory Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kwon, Youngwook; Park, Hyunsu; Choi, Yoonjae; Sim, Jincheol; Choi, Jonghyuck; Park, Seungwoo; Kim, Kyeong-Min; Choi, Changkyu; Jung, Hae-Kang; Kim, Chulwoo

被引用次数: 6

2023

A 25-Gb/s Single-Ended PAM-4 Receiver With Time-Windowed LSB Decoder for High-Speed Memory Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Yoonjae; Park, Hyunsu; Choi, Jonghyuck; Sim, Jincheol; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Sim, Changmin; Kim, Chulwoo

被引用次数: 3

2023

A Thermoelectric Energy-Harvesting Interface With Dual-Conversion Reconfigurable DC-DC Converter and Instantaneous Linear Extrapolation MPPT Method

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Inho; Jeon, Jinwoo; Kim, Hyunjin; Park, Taehyeong; Kim, Chulwoo; Jeong, Junwon

被引用次数: 3

2023

An Automotive 12 V-to-1.2 V Integrated Switched-Capacitor DCDC Converter for Improved Load Transient Response

IEEE TRANSACTIONS ON POWER ELECTRONICS

Park, Inho; Kim, Hyunjin; Park, Taehyeong; Jeong, Junwon; Kim, Chulwoo

被引用次数: 1

2023

A 4-GHz Ring-Oscillator-Based Digital Sub-Sampling PLL With Energy-Efficient Dual-Domain Phase Detector

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Choi, Yoonjae; Park, Hyunsu; Choi, Jonghyuck; Sim, Jincheol; Kwon, Youngwook; Park, Seungwoo; Sim, Changmin; Kim, Chulwoo

被引用次数: 1

2023

PAM-4 Receiver With 1-Tap DFE Using Clocked Comparator Offset Instead of Threshold Voltages for Improved LSB BER Performance

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Sim, Jincheol; Park, Hyunsu; Choi, Yoonjae; Choi, Jonghyuck; Kwon, Youngwook; Kim, Chulwoo

被引用次数: 3

2023

A 16-Gb/s NRZ Receiver With 0.0019-pJ/bit/dB 1-Tap Charge-Redistribution DFE

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Jonghyuck; Choi, Yoonjae; Park, Hyunsu; Sim, Jincheol; Kwon, Youngwook; Park, Seungwoo; Kim, Seongcheol; Kim, Chulwoo

被引用次数: 1

2023

A 15-Gb/s Single-Ended NRZ Receiver Using Self-Referenced Technique With 1-Tap Latched DFE for DRAM Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Seongcheol; Sim, Jincheol; Park, Hyunsu; Choi, Yoonjae; Choi, Jonghyuck; Kim, Chulwoo

被引用次数: 6

2023

An 88.9-dB SNR Fully-Dynamic Noise-Shaping SAR Capacitance-to-Digital Converter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lim, Chaegang; Choi, Yohan; Song, Jaeyeun; Ahn, Soonsung; Jang, Seokwon; Kim, Chulwoo

被引用次数: 8

2022

A 9-bit 500-MS/s 2-bit/cycle SAR ADC With Error-Tolerant Interpolation Technique

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Jaeyeun; Park, Yunsoo; Lim, Chaegang; Choi, Yohan; Ahn, Soonsung; Park, Sooho; Kim, Chulwoo

被引用次数: 6

2022

A 56-Gb/s PAM-4 Receiver Using Time-Based LSB Decoder and S/H Technique for Robustness to Comparator Voltage Variations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Hyunsu; Sim, Jincheol; Choi, Yoonjae; Choi, Jonghyuck; Kwon, Youngwook; Kim, Chulwoo

被引用次数: 15

2022

A Four-Phase Hybrid Step-UpDown Converter With RMS Inductor Current Reduction and Delay-Based Zero-Current Detection

IEEE TRANSACTIONS ON POWER ELECTRONICS

Park, Inho; Maeng, Junyoung; Jeon, Jinwoo; Kim, Hyunjin; Kim, Chulwoo

被引用次数: 9

2022

Analysis of a Multiwire, Multilevel, and Symbol Correlation Combination Scheme

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Choi, Jonghyuck; Choi, Yoonjae; Park, Hyunsu; Sim, Jincheol; Kwon, Youngwook; Park, Seungwoo; Kim, Chulwoo

被引用次数: 4

2022

A 15 Gb/s Non-Return-to-Zero Transmitter With 1-Tap Pre-Emphasis Feed-Forward Equalizer for Low-Power Ground Terminated Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kwon, Youngwook; Park, Hyunsu; Choi, Yoonjae; Sim, Jincheol; Choi, Jonghyuck; Park, Seungwoo; Kim, Chulwoo

被引用次数: 9

2022

A 2.4-8 GHz Phase Rotator Delay-Locked Loop Using Cascading Structure for Direct Input-Output Phase Detection

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Park, Hyunsu; Sim, Jincheol; Choi, Yoonjae; Choi, Jonghyuck; Kwon, Youngwook; Kim, Chulwoo

被引用次数: 5

2022

A 25 Gb/s Wireline Receiver With Feedforward and Feedback Equalizers at Analog Front-End

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sim, Jincheol; Lee, Yeonho; Park, Hyunsu; Choi, Yoonjae; Choi, Jonghyuck; Kim, Chulwoo

被引用次数: 10

2022

A 4.5 Gb/s/pin transceiver with hybrid inter-symbol interference and far-end crosstalk equalization for next-generation high-bandwidth memory interface

ELECTRONICS LETTERS

Yoon, Kungryun; Park, Hyunsu; Choi, Yoonjae; Sim, Jincheol; Choi, Jonghyuck; Kim, Chulwoo

被引用次数: 1

2022

A 266-3750 MHz Wide-Range Adaptive Phase-Rotator-Based All Digital DLL for LPDDR5 Controllers

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Lee, Jeewan; Choi, Yoonjae; Kim, Chulwoo

被引用次数: 0

2022

A 1.69-pJ/b 14-Gb/s Digital Sub-Sampling CDR With Combined Adaptive Equalizer and Self-Error Corrector

IEEE ACCESS

Choi, Yoonjae; Hwang, Sewook; Lee, Yeonho; Park, Hyunsu; Choi, Jonghyuck; Sim, Jincheol; Kim, Chulwoo

被引用次数: 0

2021

A Dual-Mode Continuously Scalable-Conversion-Ratio SC Energy Harvesting Interface With SC-Based PFM MPPT and Flying Capacitor Sharing Scheme

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hyunjin; Maeng, Junyoung; Park, Inho; Jeon, Jinwoo; Choi, Yohan; Kim, Chulwoo

被引用次数: 21

2021

A 1.3-4-GHz Quadrature-Phase Digital DLL Using Sequential Delay Control and Reconfigurable Delay Line

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Hyunsu; Sim, Jincheol; Choi, Yoonjae; Choi, Jonghyuck; Kwon, Youngwook; Park, Seungwoo; Park, Gyutae; Chung, Jinil; Kim, Kyeong-Min; Jung, Hae-Kang; Kim, Hyungsoo; Chun, Junhyun; Kim, Chulwoo

被引用次数: 10

2021

A 90.2% Peak Efficiency Multi-Input Single-Inductor Multi-Output Energy Harvesting Interface With Double-Conversion Rejection Technique and Buck-Based Dual-Conversion Mode

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hyunjin; Maeng, Junyoung; Park, Inho; Jeon, Jinwoo; Lim, Dongju; Kim, Chulwoo

被引用次数: 27

2021

A High-Voltage Dual-Input Buck Converter With Bidirectional Inductor Current for Triboelectric Energy-Harvesting Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Maeng, Junyoung; Park, Inho; Shim, Minseob; Jeong, Junwon; Kim, Chulwoo

被引用次数: 18

2021

30-Gb/s 1.11-pJ/bit Single-Ended PAM-3 Transceiver for High-Speed Memory Links

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Hyunsu; Song, Junyoung; Sim, Jincheol; Choi, Yoonjae; Choi, Jonghyuck; Yoo, Jeongsik; Kim, Chulwoo

被引用次数: 31

2021

A Periodically Refreshed Capacitive Floating Level Shifter for Conditional Switching Applications

IEEE TRANSACTIONS ON POWER ELECTRONICS

Maeng, Junyoung; Shim, Minseob; Jeong, Junwon; Park, Inho; Kim, Chulwoo

被引用次数: 6

2021

A Capacitively Coupled CT $\Delta\Sigma$ M With Chopping Artifacts Rejection for Sensor Readout ICs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lim, Chaegang; Choi, Yohan; Park, Yunsoo; Song, Jaegeun; Ahn, Soon-Sung; Park, Sooho; Kim, Chulwoo

被引用次数: 6

2021

A 0.99-pJ/b 15-Gb/s Counter-Based Adaptive Equalizer Using Single Comparator in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Yoonjae; Lee, Yeonho; Park, Hyunsu; Choi, Jonghyuck; Sim, Jincheol; Kwon, Youngwook; Kim, Chulwoo

被引用次数: 5

2021

A 32-Gb/s Dual-Mode Transceiver With One-Tap FIR and Two-Tap IIR RX Only Equalization in 65-nm CMOS Technology

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Song, Junyoung; Hwang, Sewook; Kim, Chulwoo

被引用次数: 0

2021

A 0.37-in. 5900PPI liquid crystal on silicon CMOS SoC using low voltage high dynamic voltage range novel pixel circuit for augmented reality micro-displays

JOURNAL OF THE SOCIETY FOR INFORMATION DISPLAY

Jeong, Minsu; Kim, Eunsu; Hong, Ook; Kim, Chulwoo

被引用次数: 2

2021

An 11-b 100-MS/s Fully Dynamic Pipelined ADC Using a High-Linearity Dynamic Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Yunsoo; Song, Jaegeun; Choi, Yohan; Lim, Chaegang; Ahn, Soonsung; Kim, Chulwoo

被引用次数: 14

2020

A Sub-fs-FoM Digital LDO Using PMOS and NMOS Arrays With Fully Integrated 7.2-pF Total Capacitance

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Maeng, Junyoung; Shim, Minseob; Jeong, Junwon; Park, Inho; Park, Yunsoo; Kim, Chulwoo

被引用次数: 7

2020

A High-Voltage Dual-Input Buck Converter Achieving 52.9% Maximum End-to-End Efficiency for Triboelectric Energy-Harvesting Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Inho; Maeng, Junyoung; Shim, Minseob; Jeong, Junwon; Kim, Chulwoo

被引用次数: 28

2020

An Efficiency-Aware Cooperative Multicharger System for Photovoltaic Energy Harvesting Achieving 14 Efficiency Improvement

IEEE TRANSACTIONS ON POWER ELECTRONICS

Jeong, Junwon; Shim, Minseob; Maeng, Junyoung; Park, Inho; Kim, Chulwoo

被引用次数: 8

2020

A High-Efficiency Charger With Adaptive Input Ripple MPPT for Low-Power Thermoelectric Energy Harvesting Achieving 21% Efficiency Improvement

IEEE TRANSACTIONS ON POWER ELECTRONICS

Jeong, Junwon; Shim, Minseob; Maeng, Junyoung; Park, Inho; Kim, Chulwoo

被引用次数: 27

2020

A DLL-Based Quadrature Clock Generator With a 3-Stage Quad Delay Unit Using the Sub-Range Phase Interpolator for Low-Jitter and High-Phase Accuracy DRAM Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yoon, Youngbog; Park, Hyunsu; Kim, Chulwoo

被引用次数: 12

2020

An Area-Efficient and Wide-Range Inter-Signal Skew Compensation Scheme With the Embedded Bypass Control Register Operating as a Binary Search Algorithm for DRAM Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yoon, Youngbog; Kim, Chulwoo

被引用次数: 0

2020

A 0.5 V 10-bit 3 MS/s SAR ADC With Adaptive-Reset Switching Scheme and Near-Threshold Voltage-Optimized Design Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Song, Jaegeun; Jun, Jaehun; Kim, Chulwoo

被引用次数: 19

2020

Near threshold voltage digital PLL using low voltage optimised blocks for AR display system

IET CIRCUITS DEVICES & SYSTEMS

Jun, Jaehun; Lee, Sangsu; Kim, Chulwoo

被引用次数: 0

2020

12-Gb/s Over Four Balanced Lines Utilizing NRZ Braid Clock Signaling With No Data Overhead and Spread Transition Scheme for 8K UHD Intra-Panel Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Yeonho; Choi, Yoonjae; Song, Junyoung; Hwang, Sewook; Bae, Sang-Geun; Jun, Jaehun; Kim, Chulwoo

被引用次数: 8

2019

A 42 nJ/Conversion On-Demand State-of-Charge Indicator for Miniature IoT Li-Ion Batteries

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Junwon; Jeong, Seokhyeon; Sylvester, Dennis; Blaauw, David; Kim, Chulwoo

被引用次数: 4

2019

Fully Integrated Low-Power Energy Harvesting System With Simplified Ripple Correlation Control for System-on-a-Chip Applications

IEEE TRANSACTIONS ON POWER ELECTRONICS

Shim, Minseob; Jeong, Junwon; Maeng, Junyoung; Park, Inho; Kim, Chulwoo

被引用次数: 13

2019

A 9 Gb/s/ch Transceiver With Reference-Less Data-Embedded Pseudo-Differential Clock Signaling for Graphics Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Song, Junyoung; Kim, Yongtae; Kim, Chulwoo

被引用次数: 4

2019

A $\Delta \Sigma$ Modulator-Based Spread-Spectrum Clock Generator with Digital Compensation and Calibration for Phase-Locked Loop Bandwidth

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Bae, Sang-Geun; Hwang, Sewook; Song, Junyoung; Lee, Yeonho; Kim, Chulwoo

被引用次数: 2

2019

A Parasitics-Induced Failure Mechanism for Transistors in the Bit-Line Sense Amplifier Region of DDP DDR3 DRAM During a CDM Event

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Lim, Dongju; Seung, Manho; Lee, Yoonsung; Lee, Seokkiu; Kim, Chulwoo

被引用次数: 0

2019

A Near-Threshold Voltage Oriented Digital Cell Library for High-Energy Efficiency and Optimized Performance in 65nm CMOS Process

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jun, Jaehun; Song, Jaegeun; Kim, Chulwoo

被引用次数: 19

2018

A 1-V 10-Gb/s/pin Single-Ended Transceiver With Controllable Active-Inductor-Based Driver and Adaptively Calibrated Cascaded-Equalizer for Post-LPDDR4 Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Song, Junyoung; Hwang, Sewook; Lee, Hyun-Woo; Kim, Chulwoo

被引用次数: 17

2018

A Spread Spectrum Clock Generator With Nested Modulation Profile for a High-Resolution Display System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jun, Jaehun; Bae, Sang-Geun; Lee, Yeonho; Kim, Chulwoo

被引用次数: 3

2018

A Low-Power Post-LPDDR4 Interface Using AC Termination at RX and an Active Inductor at TX

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yoo, Jeongsik; Lee, Yeonho; Choi, Yoonjae; Park, Hyunsu; Park, Sanghune; Kim, Chulwoo

被引用次数: 4

2018

Edge-Pursuit Comparator: An Energy-Scalable Oscillator Collapse-Based Comparator With Application in a 74.1 dB SNDR and 20 kS/s 15 b SAR ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shim, Minseob; Jeong, Seokhyeon; Myers, Paul D.; Bang, Suyoung; Shen, Junhua; Kim, Chulwoo; Sylvester, Dennis; Blaauw, David; Jung, Wanyeong

被引用次数: 53

2017

3-Gb/s High-Speed True Random Number Generator Using Common-Mode Operating Comparator and Sampling Uncertainty of D Flip-Flop

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bae, Sang-Geun; Kim, Yongtae; Park, Yunsoo; Kim, Chulwoo

被引用次数: 39

2017

A 1.62-5.4-Gb/s Receiver for DisplayPort Version 1.2a With Adaptive Equalization and Referenceless Frequency Acquisition Techniques

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Hwang, Sewook; Song, Junyoung; Lee, Yeonho; Kim, Chulwoo

被引用次数: 10

2017

A 5-GHz Subsampling PLL-Based Spread-Spectrum Clock Generator by Calibrating the Frequency Deviation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Bae, Sang-Geun; Kim, Gyungmin; Kim, Chulwoo

被引用次数: 7

2017

A 2-Gb/s/ch Data-Dependent Swing-Limited On-Chip Signaling for Single-Ended Global I/O in SDRAM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

You, Jungtaek; Song, Junyoung; Kim, Chulwoo

被引用次数: 1

2017

A 250-Mb/s to 6-Gb/s Referenceless Clock and Data Recovery Circuit With Clock Frequency Multiplier

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Ja-Young; Song, Junyoung; You, Jungtaek; Hwang, Sewook; Bae, Sang-Geun; Kim, Chulwoo

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A Scalable Bandwidth Mismatch Calibration Technique for Time-Interleaved ADCs

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An Add-On Type Real-Time Jitter Tolerance Enhancer for Digital Communication Receivers

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A 100-nW 9.1-ENOB 20-kS/s SAR ADC for Portable Pulse Oximeter

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Spin-Based Complementary Logic Device Using Datta-Das Transistors

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Rail-to-rail regulating voltage-controlled oscillator with low supply and ground noise sensitivity

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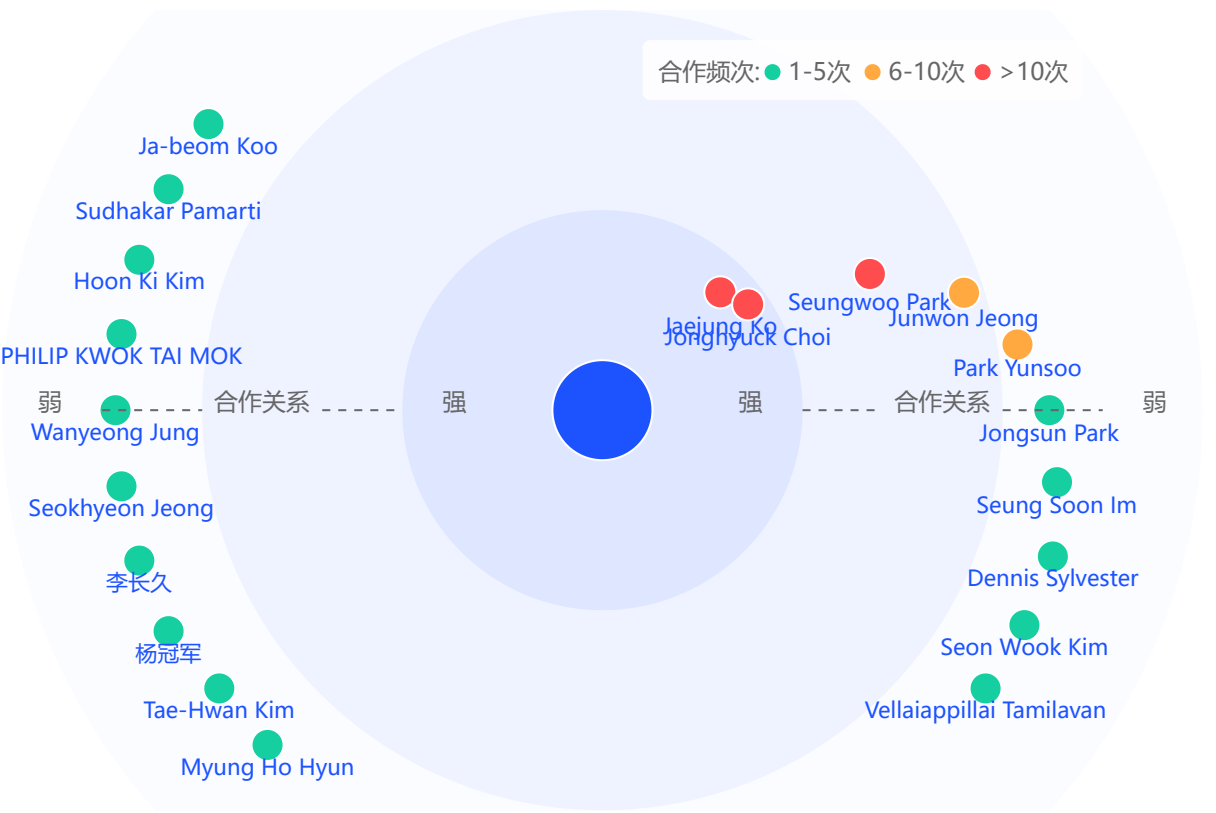
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Jaejung Ko	合作次数: 25
高丽大学	
Jonghyuck Choi	合作次数: 24
高丽大学	
Seungwoo Park	合作次数: 15

高丽大学

Junwon Jeong

合作次数: 9

高丽大学

Park Yunsoo

合作次数: 6

高丽大学

还有 30 个合作者